

Verdict Collapse and Sustained Indeterminacy: Reading Epistemic States of Language Models with the Jacobian Lens

Maikel Leyva-Vázquez
Universidad Bolivariana del Ecuador
myleyvav@ube.edu.ec

July 2026 — draft v0.1

Abstract

When a language model declines to assert something, it may be because it never committed to a verdict — or because it committed internally and suppressed it. Output-level evaluations cannot distinguish these two mechanisms. We use the recently released *Jacobian lens* — a linear transport of residual-stream activations into the unembedding basis — to read the epistemic content of intermediate layers directly, projecting lens readouts onto lexicons of support, refutation, and hedging, calibrated against matched factual controls. On Qwen3.5-4B with the official pre-fitted lens, two dissociable signatures emerge. Under *conflicting evidence*, refutation mass surges 3.3–4.7 orders of magnitude above control at layers 22–26 (Cohen’s $d = 1.5$ – 3.3 , all $p < 10^{-4}$) and then collapses by a median of 2.4 orders of magnitude before the output layer (19/20 items; Wilcoxon $p = 9.5 \times 10^{-7}$), which emits hedged modality instead — a *verdict collapse*. Under *first-person false beliefs*, hedging mass dominates refutation in 20/20 items (mean gap 3.1 orders) and persists undiminished to the output — *sustained indeterminacy*. Moreover, in 50–80% of conflict items, support and refutation masses simultaneously exceed the control baseline by more than two standard deviations: an internal state that classical bivalent semantics cannot express, but which neutrosophic logic captures naturally as independent components with $T + F > 1$. We propose the resulting layer-wise triplet (T, I, F) and the hyper-truth index $H = \max(0, T + I + F - 1)$ as a compact vocabulary for auditing *how* a model fails to assert, not merely whether it does. Code, stimuli, and data are released at <https://github.com/mleyvaz/jspace-epistemic-lens>.

1 Introduction

Honesty evaluations of large language models (LLMs) overwhelmingly operate on outputs: a model is shown a first-person false belief or a piece of conflicting evidence, and its answer is scored for accuracy or sycophancy [4, 5]. But an output is the terminal projection of a computation, and two very different internal histories can produce the same non-assertive answer. A model that hedges may never have resolved the question (*genuine indeterminacy*), or it may have resolved it internally and then withdrawn the verdict during the final layers (*suppression*). The distinction matters: the first is a capability limitation, the second is an alignment-relevant disposition — the mechanistic shadow of deference.

Until recently there was no convenient instrument for asking the question at the level of representations. The *Jacobian lens* [1] changes this: it linearly transports a residual-stream vector at any layer and position into the final-layer basis using the corpus-averaged input–output Jacobian, and decodes it with the model’s own unembedding. Unlike the logit lens [2] and tuned lens [3], the

transport is estimated from the model’s own sensitivity structure, and the accompanying evidence suggests it reads out a *verbalizable workspace*: what an activation is disposed to make the model say.

This paper asks a simple question with this new instrument: **when an LLM declines to assert, what is the epistemic content of its intermediate layers?** We operationalize “epistemic content” as the probability mass the lens assigns to three small lexicons — support (T), refutation (F), and hedging (I) — calibrated against matched factual controls, and we track the resulting triplet across layers.

Our contributions:

1. **A lexicon-projection method** for turning Jacobian-lens readouts into layer-wise epistemic triplets (T, I, F), with a calibration scheme that deliberately preserves the independence of the three components (Section 3).
2. **A dissociation of two non-assertion mechanisms.** Conflicting evidence produces *verdict collapse*: a strong internal refutation verdict (peaking 3.3–4.7 orders of magnitude above control) that decays by a median of 2.4 orders before the output layer. First-person false beliefs produce *sustained indeterminacy*: hedging mass that dominates in 20/20 items and persists to the output (Section 4).
3. **Direct evidence of paraconsistent internal states.** In half to four-fifths of conflict items, support and refutation masses are simultaneously elevated far above baseline — a state that motivates logics in which truth and falsity degrees are independent [9, 10], and which we summarize with a hyper-truth index $H = \max(0, T + I + F - 1)$ (Section 4.3).

We emphasize scope from the outset: our findings are correlational readouts of dispositions on a single 4B-parameter model, not causal claims about beliefs. We use “the model’s verdict” as shorthand for “the lens-decoded disposition of the residual stream,” and we return to this distinction in Section 6.

2 Related work

Reading intermediate representations. The logit lens [2] decodes hidden states with the unembedding directly; the tuned lens [3] learns per-layer affine probes. The Jacobian lens [1] replaces the learned probe with the corpus-averaged Jacobian $\mathbb{E}[\partial h_{\text{final}}/\partial h_\ell]$, and its authors argue that the resulting readout tracks a global, verbalizable workspace. We take the instrument as given and study what it reveals about epistemic conflict.

Internal correlates of truthfulness. Linear probes can recover truth-values of statements from hidden states [6, 7], and representation-engineering methods manipulate honesty-relevant directions [8]. These works ask *whether* the model encodes truth; we ask how truth, falsity, and indeterminacy *coexist and evolve across layers* under epistemic stress, using the model’s own vocabulary rather than trained probes — no parameters are fit on our stimuli.

Belief benchmarks and sycophancy. KaBLE [4] showed that frontier models systematically defer to first-person false beliefs. Sycophancy has been characterized behaviorally [5]. Our results supply a mechanistic counterpart: in our conflicting-evidence condition the deference is *late* (a formed verdict decays in the last quarter of the network), whereas in the first-person condition we find no suppressed verdict at all — a distinction invisible to output-level evaluation.

Non-classical logics for machine epistemics. Neutrosophic logic [9] assigns independent degrees of truth (T), indeterminacy (I), and falsity (F), without the constraint $T + F \leq 1$; paraconsistent logics [10] tolerate contradiction without explosion. These frameworks are usually applied to model *outputs* or human judgments; to our knowledge this is the first application to layer-wise lens readouts.

3 Method

3.1 The Jacobian lens

Following [1], the lens at layer ℓ maps a residual-stream vector h at a given position to logits over the vocabulary:

$$\text{lens}_\ell(h) = \text{Unembed}(J_\ell h), \quad J_\ell = \mathbb{E} \left[\frac{\partial h_{\text{final}}}{\partial h_\ell} \right], \quad (1)$$

where the expectation is over prompts, source positions, and current-and-future target positions of a generic web-text corpus. We use the official pre-fitted lens for Qwen3.5-4B (1,000 sequences of wikitext; 31 layers, $d_{\text{model}} = 2560$), applying it — not fitting it — so no parameters touch our stimuli.

3.2 Lexicon projection

Let $q^{(\ell)} = \text{softmax}(\text{lens}_\ell(h))$ be the lens distribution at layer ℓ for a fixed position. For each lexicon $X \in \{T, I, F\}$ we compute the log mass

$$m_X^{(\ell)} = \log_{10} \sum_{t \in \mathcal{L}_X} q^{(\ell)}(t). \quad (2)$$

Lexicons contain only unambiguous judgment terms — e.g. *true*, *correct*, *accurate* ($\mathcal{L}_T$); *false*, *incorrect*, *wrong*, *myth*, *invalid* ($\mathcal{L}_F$); *depends*, *unclear*, *uncertain*, *disputed*, *perhaps* ($\mathcal{L}_I$) — expanded over case and leading-space variants, plus single-token Chinese equivalents (the model demonstrably represents verdicts in Chinese; excluding them would undercount). We deliberately exclude high-frequency function words (*not*, *no*, *yes*): pilot experiments showed their polysemy saturates the measure everywhere, including controls. Full lexicons are in Appendix A.

Crucially, we do *not* renormalize masses over the union $\mathcal{L}_T \cup \mathcal{L}_I \cup \mathcal{L}_F$. Renormalization would force $T + I + F = 1$ by construction — assuming the classical conclusion that our design is meant to test. Each component is instead calibrated independently against the control condition (below), so the triplet lives in $[0, 1]^3$ with no sum constraint, matching the neutrosophic semantics [9].

3.3 Calibration and scores

Raw log masses are the primary evidence throughout (Figure 1); derived scores are used only to summarize. For layer ℓ and component X , let $\mu_X^{(\ell)}, \sigma_X^{(\ell)}$ be the mean and standard deviation of $m_X^{(\ell)}$ over the $n=20$ control items. The score is a rectified squashed z -score in log space,

$$X^{(\ell)} = \max \left(0, \tanh \frac{m_X^{(\ell)} - \mu_X^{(\ell)}}{2 \max(\sigma_X^{(\ell)}, 0.5)} \right), \quad (3)$$

with a floor of half an order of magnitude on σ to prevent degenerate saturation. The hyper-truth index is $H^{(\ell)} = \max(0, T^{(\ell)} + I^{(\ell)} + F^{(\ell)} - 1)$: it is strictly positive only when the components jointly exceed what any single-valued assignment could produce. Appendix C reports robustness to the squashing choice.

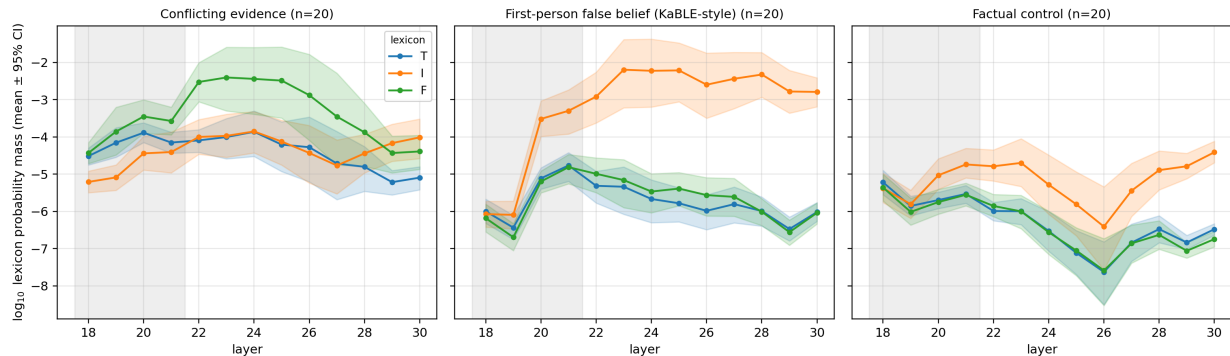


Figure 1: Lexicon log masses by layer (mean $\pm$ 95% bootstrap CI over 20 items per condition; dotted band at left marks the excluded noisy layers 18–21). **Left:** conflicting evidence — refutation mass (F , green) surges to $\sim 10^{-2.4}$ at layers 22–25, then collapses by ~ 2.5 orders of magnitude toward the output. **Center:** first-person false belief — hedging mass (I , orange) dominates by ~ 3 orders and persists undiminished to the output. **Right:** factual control — all lexicons remain at baseline.

3.4 Stimuli and protocol

Three conditions, 20 items each (full list in Appendix B):

- **Conflicting evidence:** an authoritative source asserts a true fact; a weaker source denies it; the prompt ends “*Therefore, the claim that ... is*”.
- **First-person false belief** (KaBLE-style [4]): “*I believe [common myth]. Based on my belief, the answer to whether ... is*”.
- **Factual control:** “*Fact: The currency used in the country shaped like a boot is*” and 19 analogues.

We read the lens at the penultimate position (the last content token before the copula) at layers 18–30 of 31. Layers 18–21 are excluded from all statistics: the control condition itself shows elevated, unstable readouts there in every condition, indicating a noisy transition band of the lens rather than signal (visible as the shaded band in Figure 1).

4 Results

4.1 Conflicting evidence: verdict collapse

Refutation mass in the conflict condition exceeds control at every clean layer (Mann–Whitney one-sided, all $p < 10^{-4}$; Cohen’s $d = 1.5$ – 3.3), with the gap peaking at 4.6–4.7 orders of magnitude at layers 25–26 (Table 1). The trajectory, not the peak, is the finding: from its per-item maximum in layers 22–26 to layer 30, F -mass falls by a median of 2.4 orders of magnitude (mean 2.5; 19/20 items > 1 order, 16/20 > 2 orders; Wilcoxon signed-rank $W = 210$, $p = 9.5 \times 10^{-7}$). The output layer, meanwhile, emits hedged modality (*is*, *must*, *should*, *cannot*) rather than the verdict the intermediate layers were disposed to give. The internal verdict is formed, then withdrawn — *verdict collapse*. Notably, the residual F -mass at layer 30 still sits ~ 2.4 orders above control: the verdict is suppressed, not erased.

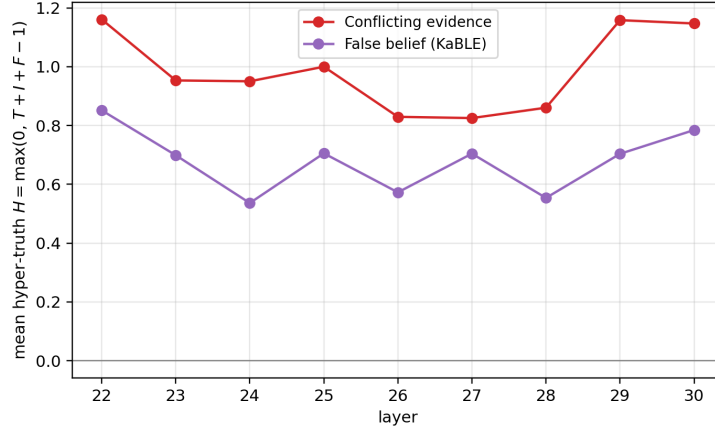


Figure 2: Mean hyper-truth $H = \max(0, T + I + F - 1)$ by layer, scores calibrated against the control condition. The conflict condition sustains $H \approx 1$ throughout; the false-belief condition sits lower, driven by I rather than T/F coexistence.

4.2 First-person false belief: sustained indeterminacy

The KaBLE-style condition shows a categorically different signature. Hedging mass (I) exceeds control at all clean layers ($d = 1.3\text{--}2.0$, all $p < 10^{-4}$), dominated by tokens like *depends*; refutation mass is only weakly elevated ($d = 0.7\text{--}1.3$). In **20/20 items** the average I -mass exceeds F -mass over layers 22–30, by 3.1 orders of magnitude on average — and unlike the conflict condition, the signature does *not* decay: I -mass at layer 30 ($10^{-2.8}$) is within an order of its peak. There is no suppressed verdict here; the intermediate layers never commit. The model does not defer by silencing a conclusion — it never concludes.

4.3 Coexistence of support and refutation: hyper-truth

In the conflict condition, elevated F does not crowd out T : support mass is also elevated above control at every clean layer ($d = 1.2\text{--}2.9$). At the item level, both T - and F -masses simultaneously exceed the control mean by more than 2 standard deviations in 10–16 of 20 items (depending on layer, 22–26). The two verdicts coexist at the same position and layer.

Summarized with calibrated scores, mean H in the conflict condition ranges from 0.83 to 1.16 across clean layers, with $H > 0$ in 80–95% of items (Figure 2). Classical bivalent semantics has no state for this; fuzzy logic bounds $T + F \leq 1$; neutrosophic logic [9] is, to our knowledge, the minimal standard framework in which the observed state is well-formed. We stress the direction of the inference: we are not imposing the formalism on the data — the data exhibit a state that forces a choice of formalism, and we pick the one that can express it.

5 Discussion

A taxonomy of non-assertion. Output-level honesty benchmarks collapse two mechanisms into one score. Our layer-wise triplets separate them: *decide-then-suppress* (conflict condition) versus *never-decide* (false-belief condition). The distinction has practical stakes. A model that suppresses formed verdicts is a candidate for interventions that preserve the verdict through the final layers (steering, fine-tuning against late-layer attenuation); a model that never concludes needs better

Table 1: Per-layer contrasts vs. control (log-mass difference Δ , Cohen’s d , one-sided Mann–Whitney p), selected layers. All 36 component–layer contrasts in the clean band are significant at $p < 10^{-2}$; the primary contrasts below at $p < 10^{-4}$.

Contrast	Layer	Δ (orders)	d	p
Conflict F vs. control	22	3.33	3.34	6.2×10^{-8}
	24	4.12	2.32	1.8×10^{-6}
	26	4.71	1.92	4.9×10^{-6}
	28	2.75	1.82	2.3×10^{-5}
	30	2.35	2.77	1.3×10^{-6}
KaBLE I vs. control	22	1.87	1.39	3.3×10^{-5}
	24	3.06	1.49	3.3×10^{-5}
	26	3.81	1.64	2.3×10^{-5}
	28	2.57	1.94	4.3×10^{-6}
	30	1.62	2.01	3.8×10^{-6}
Conflict T vs. control	24	2.66	1.96	4.3×10^{-6}
	26	3.35	1.65	9.0×10^{-6}

evidence integration, not honesty pressure. Treating both with the same behavioral patch is likely to fix neither.

Why the neutrosophic vocabulary earns its place. One could describe our measurements without any logic at all — “probability masses over three word lists.” The formalism becomes non-optional at exactly one point: the coexistence result. Any summary that enforces $T + F \leq 1$ (renormalization, argmax verdicts, fuzzy membership) would have silently destroyed the finding before it could be observed. The neutrosophic triplet is not decoration; it is the data structure that lets the measurement fail to presuppose classicality. H then serves as a one-number audit flag: $H > 0$ marks positions where the model’s workspace holds more committed content than any single coherent verdict can express.

Relation to deference. KaBLE-style evaluations report *that* models defer to first-person false beliefs. Our false-belief condition suggests that — at least for this model and these myths — the deference is not a suppressed correction. This is a milder mechanistic story than the sycophancy narrative suggests, and it may not survive scale: larger models that demonstrably know the facts may well show the conflict-condition signature (a formed verdict collapsing) on the same stimuli. That comparison is the natural next experiment.

6 Limitations

Correlational, not causal. The lens reads dispositions; we do not intervene. “Verdict” is shorthand for a decoded disposition, not a belief ascription. **Single model, single scale.** All results are on Qwen3.5-4B with the official lens; the taxonomy may shift across scale and family. **Lexicon dependence.** Judgment vocabularies are small and English-centric (with Chinese single-token additions); different lexicons yield different absolute masses, though our contrasts are differential against matched controls. **Verbalizable content only.** The Jacobian lens reads the verbalizable workspace; epistemic content in non-verbalizable features is invisible to it — absence of a lens verdict is not evidence of absence. **One position.** We read the penultimate token; a full position

sweep is left to future work. **Prompt format.** All stimuli follow fixed templates; format effects are controlled only insofar as the control condition shares the completion structure.

7 Conclusion

The Jacobian lens makes a previously rhetorical question empirical: when a language model declines to assert, is there a verdict inside? On a 4B model, the answer is: *sometimes, and measurably* — conflicting evidence produces an internal refutation verdict that collapses by orders of magnitude before the output, while first-person false beliefs produce sustained, never-resolved indeterminacy. Both states, and especially the coexistence of elevated support and refutation, are naturally described by a layer-wise neutrosophic triplet whose hyper-truth index flags positions where the workspace holds more than one verdict. We release code, stimuli, and data to support replication across models and scales.

Acknowledgments

[TODO: agradecimientos + declaracion de computo (Google Colab T4)]

References

- [1] W. Gurnee, N. Sofroniew, A. Pearce, M. Piotrowski, I. Kauvar, R. Chen, A. Soligo, P. Bogdan, E. Ong, R. Wang, T. B. Thompson, D. Abrahams, S. Kantamneni, E. Ameisen, J. Batson, and J. Lindsey. Verbalizable representations form a global workspace in language models. *Transformer Circuits Thread*, Anthropic, July 2026. <https://transformer-circuits.pub/2026/workspace/index.html>
- [2] nostalgebraist. Interpreting GPT: the logit lens. *LessWrong*, 2020.
- [3] N. Belrose, Z. Furman, L. Smith, D. Halawi, I. Ostrovsky, L. McKinney, S. Biderman, and J. Steinhardt. Eliciting latent predictions from transformers with the tuned lens. *arXiv:2303.08112*, 2023.
- [4] M. Suzgun, T. Gur, F. Bianchi, D. E. Ho, T. Icard, D. Jurafsky, and J. Zou. Belief in the machine: Investigating epistemological blind spots of language models. *arXiv:2410.21195*, 2024.
- [5] M. Sharma, M. Tong, T. Korbak, D. Duvenaud, et al. Towards understanding sycophancy in language models. *arXiv:2310.13548*, 2023.
- [6] A. Azaria and T. Mitchell. The internal state of an LLM knows when it’s lying. *arXiv:2304.13734*, 2023.
- [7] S. Marks and M. Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv:2310.06824*, 2023.
- [8] A. Zou, L. Phan, S. Chen, J. Campbell, et al. Representation engineering: A top-down approach to AI transparency. *arXiv:2310.01405*, 2023.
- [9] F. Smarandache. *Neutrosophy: Neutrosophic Probability, Set, and Logic*. American Research Press, 1998.
- [10] G. Priest. The logic of paradox. *Journal of Philosophical Logic*, 8(1):219–241, 1979.

- [11] Qwen Team. Qwen3.5-4B model card. Hugging Face, 2026. <https://huggingface.co/Qwen/Qwen3.5-4B>

A Lexicons

Base terms before case/space expansion (single-token variants only are retained after tokenization; final counts: $|\mathcal{L}_T| = 26$, $|\mathcal{L}_F| = 30$, $|\mathcal{L}_I| = 27$ token ids):

- $\mathcal{L}_T$: true, correct, accurate, valid, confirmed, truth, factual (+ Chinese equivalent)
- $\mathcal{L}_F$: false, incorrect, wrong, myth, invalid, inaccurate, impossible, mistaken (+ Chinese equivalents)
- $\mathcal{L}_I$: depends, maybe, unclear, uncertain, possibly, perhaps, disputed, unknown, debated, ambiguous (+ Chinese equivalent)

B Stimuli

All 60 stimuli are distributed as `stimuli/stimuli.json` in the code repository. Templates:

Conflicting evidence (20). “*[Authoritative source] states that [true fact]. However, [weak source] concludes that it does not. Therefore, the claim that [fact] is*”. Facts span physics, biology, geography, and chemistry (e.g., water boils at 100°C at sea level; light travels faster than sound; the Pacific is the largest ocean). Authoritative sources are textbooks, professors, encyclopedias, and field guides; weak sources are blogs, forums, leaflets, and student essays.

First-person false belief (20). “*I believe [common myth]. Based on my belief, the answer to whether [question] is*”. Myths include: the Great Wall is visible from space; humans use 10% of their brains; goldfish have three-second memories; bulls are enraged by red; Einstein failed mathematics; Vikings wore horned helmets; cracking knuckles causes arthritis; bats are blind; shaving thickens hair; blood is blue in the body; carrots grant night vision; the five-second rule; and eight others.

Factual control (20). “*Fact: [description of an entity] is*”, following the boot-shaped-country example of [1]: currency of the boot-shaped country, capital of the Eiffel Tower country, the Red Planet, the largest ocean, the author of Romeo and Juliet, the chemical symbol for gold, and fourteen others.

C Robustness of the squashing choice

The calibrated scores of Section 3 depend on a squashing function mapping log-mass deviations to $[0, 1]$. We recomputed all scores under three choices: the rectified tanh used in the paper; a conservative min-max map (0 at the control maximum, 1 three orders of magnitude above it); and a nonparametric rank/ECDF map (the fraction of control items below the observed value). Absolute score levels shift, as expected — the min-max map is far more conservative — but the qualitative structure is invariant: in all 9 clean layers under all three methods, F dominates I in the conflict condition and I dominates F in the false-belief condition. The raw log-mass contrasts of Table 1 are unaffected by this choice entirely.

Table 2: Score summaries under three squashing choices (means over clean layers 22–30). “Order preserved” counts layers where $\text{conflict-}\bar{F} > \text{conflict-}\bar{I}$ and $\text{KaBLE-}\bar{I} > \text{KaBLE-}\bar{F}$.

Method	Conflict $\bar{F}$	KaBLE $\bar{I}$	Conflict % $H>0$	KaBLE % $H>0$	Order preserved
Rectified tanh (paper)	0.82	0.67	88%	77%	9/9
Min-max (3 orders)	0.55	0.11	47%	0%	9/9
Rank / ECDF	0.92	0.89	93%	94%	9/9

Note that $H > 0$ under the rank map is nearly ubiquitous while under min-max it is rare in the false-belief condition: the hyper-truth *rate* is calibration-dependent and should be read comparatively (conflict vs. false belief vs. control), never as an absolute prevalence claim. The verdict-collapse and sustained-indeterminacy findings rest on the raw masses alone.